6.2 Radicals and Rational Exponents

Name _____ Period ____ Date _____

Algebra I 6.2 · Student sheet

I can…

- find nth roots.
- rewrite expressions between radical form and rational exponent form.
- evaluate expressions with rational exponents.

Vocabulary · fill in the missing word

nth root — if $b^n = a$, then b is an nth root of a, _____ $\sqrt[n]{a}$

Index — _____

Rational exponent — $a^{\frac{1}{n}}$ = _____ = $(\sqrt[n]{a})^m$

Part 1 · Exponent detective

$5^{1/2} \cdot 5^{1/2} = 5^1 = 5$. So $5^{1/2}$ is the number that times itself makes 5. That number is _____

$8^{1/3} \cdot 8^{1/3} \cdot 8^{1/3} = 8^1 = 8$. So $8^{1/3}$ = _____

Expression	$9^{1/2}$	$16^{1/2}$	$8^{1/3}$	$27^{1/3}$	$16^{1/4}$
Value					

Rule: $a^{1/n}$ means _____

Part 2 · Notes and examples (fill in each note, then work the example)

nth root: $\sqrt[n]{a}$ is a number whose _____ power is a. The n is called the _____.

Rational exponents: $a^{1/n}$ = _____ $a^{m/n}$ = _____

Example 1

Rewrite each expression in radical form.

a. $10^{\frac{1}{2}}$ b. $11^{\frac{1}{5}}$

Real nth roots: If n is odd, a has _____ real nth root. If n is even and $a > 0$, a has _____ real nth roots.

Example 2

Find the indicated real nth root(s) of a.

a. $n = 3, a = -27$ b. $n = 4, a = 16$

Example 3

Evaluate each expression.

a. $\sqrt[3]{(-64)}$ b. $-\sqrt[4]{16}$

Reading a^(m/n): In a^(m/n), the denominator tells the _____ and the numerator tells the _____.

Example 4

Evaluate each expression.

a. $16^{(\frac{3}{4})}$ b. $27^{(\frac{2}{3})}$ c. $32^{(-\frac{2}{5})}$

Example 5

A cube has a volume of 1331 cubic feet. Find the length of each side.

Part 3 · Partner practice

Partner 1

Evaluate $64^{(\frac{2}{3})}$.

Partner 2

Evaluate $81^{(-\frac{1}{4})}$.